

CONSTRUCTION OS

Master Product Requirements Document

Enterprise SaaS · Construction Operations · Company Network · Professional Identity

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Private by design. Configurable by subscription. Connected when collaboration is required.

Document Control

Version	Date	Prepared for	Purpose	Status
1.0	02 Aug 2026	Senior full-stack development teams	Technical estimation, architecture and delivery planning	Approved product baseline

This document consolidates the complete product direction and decisions confirmed during discovery. Items marked “Open” require resolution during technical inception; they do not invalidate the current scope.

How to Use This Document

- Product owners use Sections 1–15 to validate scope and operating rules.
- UX/UI leads use Sections 16–18 and the visual-reference appendix.
- Backend and platform engineers use Sections 19–25.
- QA and delivery leads use Sections 26–29.
- The development team must return effort, assumptions, exclusions, risks and phased pricing against the roadmap in Section 29.

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1. Product Vision and Executive Summary

Construction OS is a premium, private, multi-tenant operating system for construction companies. Each Company Owner and CEO should experience the product as software built exclusively for their organisation, while shared network capabilities remain embedded and appear only when needed for joint ventures, tendering, recruitment or verified-company collaboration.

Primary promise	One secure source of truth for projects, people, contracts, finance, contractors, documents, stock, approvals and decisions.
Primary buyer	Construction company CEO, Company Owner, developer, holding group or enterprise administration team.
Core users	Executives, Company Admins, HR, Finance, Legal, Architects, Engineers, Project Managers, Sales, Procurement, Site and Stock Managers.
External users	Contractors, clients, consultants and professionals with restricted portal access.
Commercial model	Modular SaaS priced by licensed users, enabled modules, storage and enterprise add-ons.
Differentiator	Private company OS with cross-company JV collaboration, company/professional identities, structured accountability and optional Digital Twin navigation.

Product Principles

- Private-first: company data is isolated and never exposed by default.
- One identity, many contexts: a person can hold company roles and an independent professional profile without separate accounts.
- Everything is traceable: every entity has a permanent ID, timeline, comments, relationships and audit history.
- Approval before authority: financial, legal, contractual, design and stock actions follow configurable approval chains.
- Premium clarity: role-specific fixed dashboards, light visual design, controlled information density and contextual help.
- Connected only when required: JV, tenders, contractor portals, client portals and business directory are integrated capabilities, not a social-network home screen.

2. Confirmed Product Decisions

Decision area	Confirmed direction
Platform	Multi-company SaaS with cross-company collaboration for Joint Venture projects.
JV visibility	Fully configurable per module and per record: private to one company, shared with all JV partners, or shared with selected users.
Subscription	Modular pricing by licensed users, modules, storage and premium options.
Data ownership	Each company owns its data. Platform support access is temporary, company-approved, time-limited and fully logged.
Customization	Subscription-based limited customization. New modules are built only by the Construction OS team as a paid service.
Database	Hybrid: shared multi-tenant database for standard customers; dedicated database option for enterprise customers.
Governance	One Company Owner and unlimited Company Admins. Only the Owner manages subscription, ownership transfer, deletion and support access.
Deletion	Owner requests deletion; company enters seven-day standby/read-only state; Platform Main Admin provides final approval.
Initial integrations	WhatsApp notifications and time tracking; optional external cloud storage connections.
Licensing	Only login-enabled users consume paid licenses; non-login labourers and other employees may remain in HR without licenses.
Languages	English and Albanian.
Authentication	Username and password. Accounts are created only by Company Admins; no public self-service access to company workspaces.
Files	Files committed into Construction OS are the official final controlled versions; external cloud storage may be connected.
Drawing revisions	Full revision control with code, description, approvals, signatures, history, notifications and linked-task updates.
Templates	Official platform templates plus company-created templates according to subscription.
Notifications	In-app, email and WhatsApp; each user chooses channels, subject to company defaults and critical-event rules.
Theme	Light mode only; professional minimal design with dark-grey accents and refined motion.
Dashboards	Fixed by role; Finance, HR, Architecture, CEO and other roles receive purpose-built dashboards.
Search	Smart global search. Restricted results are visible as locked; users may request permission.
Saving	Manual save for records; Universal Draft Mode before submission.
Offline	Full offline creation and update for supported field workflows, with encrypted local storage, sync and conflict handling.

Decision area	Confirmed direction
Digital Twin	Premium enterprise module linking model objects to tasks, drawings, progress, inspections, contracts, stock and timelines.
Company profiles	Every company has a public/private profile, parent/child structure, public projects, contacts and verified trust score.
Professional profiles	Architects, engineers and other professionals can create profiles, portfolios and job applications; HR can search and recruit.
Landing page	Login-first, flat premium page with central sign-in card, "Apply for sign up", and About menu containing Request Demo and Security.

3. Product Architecture: One Platform, Three Embedded Experiences

The product is marketed and presented as one private Construction OS. Network and career capabilities are never shown as three separate products. The user’s home, navigation and permissions are determined by identity, company membership, role and subscription.

Company Workspace	The private operating environment purchased by a company. Includes operational modules, dashboards, projects and governance.
Business Collaboration Layer	Company directory, tenders, verified profiles, JV workspaces and secure cross-company communication, shown only in relevant modules.
Professional Identity Layer	A persistent Construction Passport for professionals, with CV, portfolio, certifications, skills, jobs, invitations and verified experience.

Workspace Switching

- A user may belong to one or more companies, a parent group and JV projects.
- The same login can open a professional profile or a permitted company workspace.
- Company dashboards remain private and role-specific; professional profile functionality is secondary once the person joins a company.
- Contractor and client users see restricted portals rather than the full company OS.

Premium Positioning Rules

- The CEO landing and dashboard must display the company identity, not a marketplace feed.
- Network features are surfaced as Business Directory, Tenders, Recruitment, Public Profile or Joint Venture tools.
- No social feed appears in the default executive workspace.
- Companies may disable public profiles, job visibility, tenders or directory participation subject to subscription and platform policy.

4. Multi-Tenant SaaS, Data Isolation and Joint Ventures

TEN-001 Strict tenant isolation [Must]

All tenant-owned data must be scoped by tenant identifier at application, service, database, cache, search-index, background-job and file-storage levels.

TEN-002 Hybrid database model [Must]

Standard customers may share logically isolated databases. Enterprise customers may purchase a dedicated database and migrate without data loss.

TEN-003 Cross-tenant collaboration objects [Must]

JV projects and shared tender records must use explicit collaboration entities rather than weakening tenant isolation.

TEN-004 Record-level visibility [Must]

Every shareable record supports owner tenant, permitted tenants, selected users, permitted roles and access expiry.

TEN-005 Temporary support access [Must]

Platform staff gain no implicit access. Support access requires owner approval, reason, duration, scope and immutable audit log.

Joint Venture Project Setup

Step	Actor	Action	System result
1	Company A Owner/Admin	Creates JV workspace and invites Company B.	Pending JV invitation is created.
2	Company B Owner/Admin	Reviews terms and accepts.	JV participant tenant is linked.
3	JV Admins	Assign members and module access.	Only selected users can enter the JV project.
4	Record owner	Sets each record as private, shared to all JV parties or shared to selected users.	Visibility policy is enforced across UI, API, search and exports.
5	Approvers	Complete cross-company approvals.	Audit trail records company, person, action and decision.

Acceptance Criteria

- ☐ A user from Company B cannot discover Company A private records through API, search, notifications, exports or AI.
- ☐ A shared JV invoice can be viewed only by specifically assigned users from both companies.
- ☐ Revoking a company or user removes future access while preserving historical audit records.

5. Company, Group, Subsidiary and Branch Structure

- Every company has its own Construction OS profile and tenant identity.
- A parent company can display and govern child companies while preserving legal and data separation.
- Parent profiles show group contact information, child companies and optionally shared public projects.
- The system supports acquisitions and mergers while preserving historical ownership, visibility and audit records.
- Companies may have unlimited offices/branches. Albania-specific registered “Degë” status must be supported in company legal records.
- Branch records can contain address, local contacts, employees, projects, warehouses, equipment, working hours and local administrators.
- Resource lending between unrelated companies is not supported. Collaboration occurs through explicit JV or contract relationships.

Parent Company	Group-level identity, consolidated oversight and child-company navigation.
Child Company	Independent company workspace with its own users, projects, contracts and permissions.
Branch/Office	Operational location within one company; may be marked as legally registered branch where relevant.
Merger/Acquisition	Controlled ownership transition preserving source history and access boundaries.

6. User Personas and Account Types

Persona	Primary responsibility
Platform Main Admin	Operates the SaaS, verifies companies, approves final company deletion, manages plans and temporary support sessions.
Company Owner	Owns company workspace, subscription, ownership transfer, deletion requests and support authorization.
Company Admin	Creates users and credentials, assigns departments/positions/permissions, configures company and manages all internal modules.
CEO / Director	Executive oversight of projects, finance, approvals, risks, departments and subsidiaries.
Project Manager / Task Manager	Coordinates projects, schedules, dependencies, contributors, approvals and delay accountability.
Architect	Drawings, design packages, revisions, RFIs, approvals, tasks, decisions and floor-plan requests.
Engineer	Technical specifications, calculations, inspections, submittals, RFIs and coordination.
HR	Employee records, attendance, absence reports, recruitment, onboarding, performance, training and certifications.
Finance	Budgets, spending approvals, invoices, payments, cash flow, contractor balances and reports.
Legal	Contracts, claims, notices, penalties, disputes, legal fees, compliance and signed documents.
Sales / Commercial	CRM, unit inventory, client requests, floor-plan tasks, customisation approvals and sales packages.
Procurement	Suppliers, tenders, quotations, purchase orders, deliveries and vendor performance.
Stock / Quantity Manager	Warehouse control, daily stock confirmation, movements, tagged recipients and physical counts.
QA/QC	Inspections, non-conformities, punch lists, approvals and quality evidence.
HSE	Safety inspections, incidents, permits, risk assessments and training.
Contractor Representative	Restricted access to own contracts, tasks, meetings, penalties, invoices, documents and comments.
Client	Configurable collaboration portal for progress, approvals, variations, requests, meetings and signatures.
View-only / Trainee	Can navigate and read assigned information without creating, uploading, approving or changing records.
Professional Applicant	Independent profile, CV, portfolio, certifications, job search, applications, messages and company invitations.

7. Baseline Role and Permission Matrix

Module/Action	Owner	Admin	CEO	PM	Architect	Engineer	HR	Finance	Legal	Sales	Stock	Contractor	Client	Viewer
Company settings	Owner/Admin	Admin	Read	No	No	No	No	No	No	No	No	No	No	Read
User creation	Owner/Admin	Create	No	No	No	No	No	No	No	No	No	No	No	No
Projects	Full	Full	Full	Manage	Assigned	Assigned	Read	Finance view	Legal view	Sales view	Stock view	Assigned only	Assigned only	Read assigned
Tasks	Full	Full	Full	Manage	Create/Contribute	Create/Contribute	Assigned	Assigned	Assigned	Create/Contribute	Assigned	Assigned	Comment/Approve	Read
Finance	Full	Full	Full	Project view	Restricted	Restricted	Payroll only	Manage	Legal-linked	Sales-linked	Expense submit	Own payments	Configured	No
HR	Full	Full	Summary	Team view	Own profile	Own profile	Manage	Payroll-linked	Restricted	Own profile	Own profile	Own profile	No	No
Contracts	Full	Full	Full	Project contracts	Assigned docs	Technical docs	Employment only	Financial terms	Manage	Sales contracts	PO-linked	Own contracts	Configured	Read assigned
Audit logs	Full	Full	No by default	No	No	No	No	No	No	No	No	No	No	No
Company network	Full	Full	Read	Tender access	Read	Read	Recruitment	Financial due diligence	Legal due diligence	Business development	Supplier view	Own company	Own company	Read public

8. Authentication, Accounts and Company Administration

Controls identity, credentials, company membership, positions, permissions and account lifecycle. Only Company Admins or Owners create internal company accounts.

Primary actors	Company Owner, Company Admin, Platform Main Admin, All users
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Create user profile from inside Company Administration.
- Assign department, position, team, manager, projects and access mode.
- Generate username and temporary password; require password change on first login if enabled by policy.
- Assign standard role and optional individual permission overrides.
- Set View-only mode for onboarding or training periods.
- Activate, suspend, archive, revoke access or reset password.
- Show licensed-seat use and non-login employee count separately.
- Maintain employee account and independent professional identity as separate but linked records.

Business Rules

AUT-001 Admin-only creation [The Create User action is available only to Company Owner/Admin.]

Internal company accounts cannot self-register.

AUT-002 Credential ownership [Professional identity remains owned by the individual; company records remain company property.]

The company controls workspace credentials and can reset or revoke them.

AUT-003 View-only onboarding [View-only users cannot create tasks, upload, approve or edit.]

A new user may be limited to viewing assigned records until the Admin changes access mode.

AUT-004 One account, multiple roles [Permissions are evaluated in the active workspace.]

A user may have different roles in different company/JV workspaces.

Key Workflows

Create and Assign Architect Account

Step	Actor	Action	System result
1	Company Admin	Selects Create User and enters name, username, contact details and optional professional-profile link.	Draft user is created.
2	Company Admin	Assigns Architecture department, Architect position, manager, projects and access mode.	Role permissions are previewed.
3	Company Admin	Generates temporary credentials and confirms creation.	Account becomes active and invitation is sent.
4	Architect	Signs in with supplied username/password and changes temporary	Architect dashboard opens with only permitted modules.

Step	Actor	Action	System result
		password if required.	

Core Data

Field / Entity	Type	Required	Notes
User ID	UUID + display code	Yes	Permanent immutable identifier
Username	String	Yes	Unique within platform or configured namespace
Password hash	Secure hash	Yes	Never stored or logged in plain text
Department/Position	Reference	Yes	Controls dashboard and baseline permissions
Manager	User reference	Conditional	Used for approvals and escalations
Access mode	Enum	Yes	Standard, View-only, Suspended, Archived

Notifications

- Account created
- Credentials issued
- Role changed
- Access suspended/revoked
- Password reset
- Project access changed

Acceptance Criteria

- ☐ Admin creates an Architect account and the Architect can sign in with generated credentials.
- ☐ The Architect cannot access Finance unless an explicit permission is granted.
- ☐ View-only mode blocks all write actions and APIs.

9. Subscription, Licensing and Company Lifecycle

Manages modular SaaS plans, licensed seats, storage, premium options, dedicated database upgrades and controlled company deletion.

Primary actors	Company Owner, Company Admin, Platform Main Admin, Billing team
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Price by active licensed users, modules, storage and premium add-ons—not total employee count.
- Allow unlimited non-login employee records subject to plan policy.
- Activate or deactivate modules by subscription entitlements.
- Offer branding, dedicated database, Digital Twin, AI, advanced customization and custom development as paid options.
- Track plan, renewal, usage, storage and invoice history.
- Support seven-day deletion standby with cancellation and final platform approval.

Business Rules

SUB-001 Licensed user rule [Non-login labourers do not consume seats.]

Any user who can log into a company workspace consumes a licensed seat unless a special portal license applies.

SUB-002 Deletion safeguard [Deletion request enters seven-day standby; Platform Main Admin approves final deletion.]

Company Owner cannot instantly delete a tenant.

SUB-003 Module entitlements [Existing data remains retained according to contract and can be exported.]

Disabled modules must be inaccessible through UI and API.

Key Workflows

Company Deletion

Step	Actor	Action	System result
1	Company Owner	Submits deletion request and confirms reason.	Workspace enters standby/pending deletion.
2	System	Notifies Owner, Admins and Platform Main Admin; makes workspace read-only according to policy.	Seven-day countdown begins.
3	Company Owner	Cancels request during cooling-off period if needed.	Workspace is restored.
4	Platform Main Admin	Reviews after seven days and confirms or rejects.	Tenant is archived/deleted per retention policy and export obligations.

Notifications

- Seat threshold reached
- Storage threshold reached
- Module activation/deactivation
- Renewal due
- Deletion requested/cancelled/approved

Acceptance Criteria

- ☐ A 50-seat plan blocks creation of the 51st licensed login unless the plan is upgraded.
- ☐ Deleting a company cannot complete before the seven-day period and platform approval.

10. Human Resources, Attendance and Recruitment

Provides complete workforce control for login users and non-login employees, including attendance, absences, recruitment and professional profiles.

Primary actors	HR, Company Admin, Supervisors, Employees, Professional Applicants
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Employee profile, contract, position, salary metadata, manager and project assignments.
- Daily attendance: present, late, missing, sick, vacation, business trip, remote, training or unexcused.
- Superiors report missing workers for a specific day; HR reviews and records outcome.
- Leave requests, approvals, balances and calendar integration.
- Certifications, licenses, training, expiry reminders and performance history.
- Recruitment pipeline: job, applicants, screening, interviews, technical review, offer and hire.
- Professional Construction Passport with CV, portfolio, skills, certificates, languages, experience, availability and visibility controls.
- AI-assisted matching by skill, experience, certification, availability and workload.

Business Rules

HUM-001 Supervisor absence reporting [HR receives an immediate notification and may request evidence.]

Missing workers are recorded by their direct superior or authorised site lead.

HUM-002 Employee history [Company-confidential entries are not exported to a public professional profile.]

Employment, absence, project, performance, warning and training events remain in the employee timeline.

HUM-003 Professional visibility [Company confidential data is excluded.]

Professionals choose private, company-only, parent-group, JV or verified-company visibility.

HUM-004 Hire conversion [Role and credentials are still created by Company Admin.]

A successful applicant can become a company user or non-login employee without duplicate data entry.

Key Workflows

Report Missing Worker

Step	Actor	Action	System result
1	Supervisor	Selects employee, date, status/reason and optional evidence.	Absence report is created.
2	System	Notifies HR and updates attendance as pending review.	Employee and project timelines are updated.
3	HR	Approves, corrects or rejects the entry.	Final attendance record and payroll impact are stored.

Recruit Professional

Step	Actor	Action	System result
1	HR	Publishes job and defines skills, location and requirements.	Job appears in authorised career search.

Step	Actor	Action	System result
2	Professional	Applies using Construction Passport.	Application enters pipeline.
3	HR/Reviewers	Screen, interview and score candidate.	Decision and notes are permission-controlled.
4	HR/Admin	Marks hired and requests account creation.	Employee profile is linked; Admin creates company credentials.

Notifications

- Missing worker reported
- Leave request submitted/approved/rejected
- Certification expiring
- Candidate applied
- Interview scheduled
- Offer issued

Acceptance Criteria

- ☐ A supervisor can report a missing worker for one day and HR can approve it.
- ☐ A professional can apply without accessing private company modules.
- ☐ Hiring does not expose prior employer confidential data.

11. Projects, Programmes and Progress Control

Provides the central project workspace and company-wide oversight of schedule, scope, budget, progress, risks, documents and participants.

Primary actors	CEO, Project Manager, Architects, Engineers, Finance, Legal, Contractors, Clients
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Project metadata: client, location, budget, contract value, start/finish dates, project manager and team.
- Hierarchical work breakdown: project, building, block, floor, zone, unit, package and discipline.
- Phase progress: structure, facade, MEP, interior, landscape and configurable packages.
- Planned versus actual progress, days ahead/behind, health and forecast completion.
- Gantt schedule, dependencies, milestones, critical path and resource assignments.
- Project timeline, activity, meetings, documents, contracts, finances, risks and issues.
- Public project profile optionally exposed through company profile without confidential information.
- Project templates for residential, hotel, villa, commercial, hospital, school, infrastructure, industrial and mixed-use projects.

Business Rules

PRO-001 Calculated progress [Manual overrides require a reason, approver and audit entry.]

Project progress should derive from approved tasks, installed quantities, milestones and inspections.

PRO-002 Project access [External participants see only approved/shared records.]

Membership and role determine module access inside each project.

PRO-003 Archiving [Must preserve timelines and audit logs.]

Archived projects become read-only, searchable, restorable and downloadable as an authorised archive package.

Key Workflows

Create Project from Template

Step	Actor	Action	System result
1	Admin/Project Director	Selects template, legal entity, branch, client, project type and dates.	Project draft is created.
2	Project Manager	Reviews packages, tasks, workflows, folders and team.	Project configuration is adjusted.
3	Approver	Approves project activation if required.	Workspace becomes active and invitations are sent.

Notifications

- Project created/activated
- Milestone due
- Project delayed
- Progress override
- Project archived/restored

Acceptance Criteria

- ☐ CEO sees accurate project health without opening each project.

CONSTRUCTION OS / MASTER PRD

- ☐ Project progress cannot be silently overwritten.
- ☐ A complete authorised archive can be downloaded.

12. Tasks, Contributions, Collaboration and Decisions

The task engine is the operational core. It supports individual assignments, departmental contributions, public collaboration, voting, dependencies, approvals and mandatory delay accountability.

Primary actors	All licensed users, Contractors, Clients where permitted, Task Managers, Superiors
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Any authorised user may create a task.
- Creator assigns main responsible person, contributors, reviewers, approvers, watchers and departments.
- Each contributor receives a separate required deliverable and status.
- Required file formats, signatures, checklists and metadata can block completion.
- Task visibility: Private, Department Public or Company Public.
- Visible users can follow, comment, suggest improvements or request to join; Task Manager/Admin approves membership.
- Decision Mode supports options, discussion, evidence, informational votes and final superior decision.
- Parent/subtasks, dependencies, recurrence, priority, deadlines, estimated hours and linked records.
- Completion notifies all involved users and writes to relevant timelines.

Business Rules

TAS-001 Mandatory overdue explanation [Fields include reason, progress, new forecast, support needed, cost impact and dependency impact.]

When a deadline passes, the assignee or assigned contributor must submit a delay explanation before further progress updates.

TAS-002 Manager assessment [Must]

The Task Manager/creator must record accepted/rejected reason, impact, corrective action and revised deadline.

TAS-003 Escalation [Must]

Missing delay explanations escalate through superior, department head, project manager and director according to configurable timing.

TAS-004 Main-task completion [Must]

A parent task cannot complete until all mandatory contributions are approved.

TAS-005 Decision authority [Must]

Everyone permitted may vote and comment; votes are advisory. The designated superior/decision owner records the final decision and justification.

TAS-006 Public collaboration [Must]

Public task participants cannot edit until their join request is approved.

Key Workflows

Facade Multi-Department Task

Step	Actor	Action	System result
1	Creator	Creates "New Facade" and assigns Legal, Engineering and Architecture contributions.	Contributors receive required deliverables.
2	Legal	Uploads signed contract and annexes.	Legal contribution enters review.

Step	Actor	Action	System result
3	Engineering	Uploads specifications/calculations.	Engineering contribution enters review.
4	Architect	Uploads DWG, signed PDF, revision and material schedule.	Architecture contribution enters review.
5	Approvers	Approve or request revisions per contribution.	Parent task progress updates.
6	Task Manager	Closes after all required contributions are approved.	All members are notified; project timeline is updated.

Decision Task

Step	Actor	Action	System result
1	Creator	Creates question such as "Aluminium or steel?" and defines options, voters and decision owner.	Discussion opens.
2	Participants	Vote, comment, upload evidence and propose alternatives.	Advisory result is visible according to settings.
3	Decision Owner	Selects final option and records mandatory reasoning.	Decision is locked and linked tasks/documents update.

Core Data

Field / Entity	Type	Required	Notes
Task ID	Code	Yes	Example TSK-2026-000582
Visibility	Enum	Yes	Private, Department Public, Company Public, JV custom
Contributors	Many-to-many	Conditional	Per-person deliverables and status
Decision options	Collection	Conditional	Used in Decision Mode
Delay record	Entity	Conditional	Immutable history of each overdue cycle

Notifications

- Assigned/removed
- Mentioned
- Join requested/approved
- Submission uploaded
- Approval/rejection
- Due soon
- Overdue
- Delay explanation missing
- Completed

Acceptance Criteria

- ☐ The parent facade task cannot complete while Legal is unapproved.
- ☐ An overdue contributor is blocked from updating progress until a delay explanation is submitted.
- ☐ A superior may decide against the majority but must record a reason.

13. Timelines, Comments and Object Relationships

Every entity has a complete digital history and collaboration layer. Timelines and related items connect the entire operating system.

Primary actors	All authorised users
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Every object receives a permanent unique ID and immutable timeline.
- Timeline events cross-link user, company, project, task, document and related objects.
- Enterprise comments support rich text, mentions, files, images, voice notes, screen recordings and annotations.
- Comments may be classified as Decision, Question, Issue or Information.
- Discussions may be resolved/reopened and converted into tasks.
- Related Items panel automatically links records and allows authorised manual links.
- Edit history is retained for comments and relationship changes.

Business Rules

TIM-001 Timeline integrity [Corrections are represented as new events.]

Business events may not be deleted by normal users.

TIM-002 Cross-link security [Must]

A visible timeline event must not reveal confidential field values from a restricted related record.

TIM-003 Comment authority [Must]

Comment visibility follows parent-record visibility and explicit thread permissions.

Notifications

- Mention
- Reply
- Discussion resolved/reopened
- Comment converted to task

Acceptance Criteria

- ☐ Opening a contractor shows all authorised contract, payment, meeting and penalty events chronologically.
- ☐ A restricted contract may appear as a locked relation without exposing its content.

14. Notifications and Communication

Delivers actionable event notifications through in-app, email and WhatsApp with user choice and company governance.

Primary actors	All users, Company Admin
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- In-app notification centre with unread, priority, module and project filters.
- User-configurable channel preferences by event category.
- Company-defined defaults and mandatory channels for critical events.
- Daily/weekly digest options and quiet hours.
- Notification links deep-link to the permitted record and action.
- Delivery, read and acknowledgement status where legally relevant.

Business Rules

NOT-001 Permission-aware delivery [Must]

A notification cannot expose a record after access is revoked.

NOT-002 Critical event policy [Should]

Company can require WhatsApp/email for defined critical events.

NOT-003 No duplicate noise [Should]

Events should be grouped and de-duplicated where possible.

Acceptance Criteria

- ☐ A task completion notification reaches every included member through their selected channels.
- ☐ A user removed from a task no longer receives future task notifications.

15. Documents, Drawings, Revisions, RFIs and Submittals

Provides controlled document storage, versioning, approvals, design coordination and official final-file governance.

Primary actors	Architects, Engineers, Document Controllers, Project Managers, Contractors, Clients
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Searchable project/company folder structures with permissions.
- Universal Draft Mode; drafts visible only to author and selected collaborators until submitted.
- Final official version is committed into Construction OS.
- Revision code, description, author, reviewers, approvers, signatures and status.
- Previous versions preserved and downloadable according to permissions.
- RFIs with question, assignee, response, due date and status.
- Submittals for materials, technical data, certifications and approvals.
- PDF/drawing annotation, approval stamps, watermarks and digital signatures.
- Optional external storage import/connection; no uncontrolled two-way editing of official final records.

Business Rules

DOC-001 Official file rule [Must]

Dropping/importing a file into Construction OS as final creates the controlled official version.

DOC-002 Revision supersession [Must]

Superseded drawings remain available but clearly marked and cannot be mistaken for current.

DOC-003 Approval reset [Must]

A material content revision triggers reapproval according to workflow rules.

DOC-004 Required formats [Must]

Tasks can require DWG, PDF, RVT, IFC, XLSX or other configured formats.

Key Workflows

Drawing Revision Approval

Step	Actor	Action	System result
1	Architect	Uploads revision in Draft Mode with code and description.	Only draft collaborators can view.
2	Architect	Submits revision.	Parallel/sequential approval workflow begins.
3	Reviewers	Approve, reject or request revision with comments/annotations.	Status and timeline update.
4	Final Approver	Approves final.	Revision becomes current official version; affected users/tasks are notified.

Notifications

- Document submitted

CONSTRUCTION OS / MASTER PRD

- Revision superseded
- RFI assigned/answered/overdue
- Submittal approved/rejected
- Signature required

Acceptance Criteria

- ☐ A submitted revision preserves the previous revision and clearly marks current status.
- ☐ A contractor cannot download a private internal drawing.
- ☐ Required DWG and signed PDF block facade contribution completion until present.

16. Meetings and Minutes of Meeting

Captures every contractor/client/internal meeting, decisions, attendance, action items and legally traceable approvals.

Primary actors	Project Managers, Contractors, Clients, Legal, All invitees
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Meeting title, project, date/time, location/online link, organiser and attendees.
- Agenda topics with discussion, decision, responsible person and due date.
- Action items convertible to tasks with direct links to MoM.
- Internal review, contractor/client review, comments, correction requests and digital signatures.
- Final approved MoM locked and archived with timeline entry.
- Every meeting with a contractor must have a Minutes of Meeting record.

Business Rules

MEE-001 Mandatory contractor MoM [Must]

A contractor meeting cannot be marked complete without a MoM draft or explicit cancelled/no-meeting status.

MEE-002 Final lock [Must]

Signed final MoM is immutable; corrections require a new revision/addendum.

MEE-003 Action linkage [Must]

MoM action items retain meeting reference when converted to tasks.

Key Workflows

Contractor Meeting

Step	Actor	Action	System result
1	Organiser	Creates meeting, agenda and attendees.	Invitations are issued.
2	Organiser/Recorder	Records discussions, decisions and action items.	MoM draft is created.
3	Internal reviewers	Review and approve draft.	Contractor review is requested.
4	Contractor	Approves, comments or requests correction.	Response is logged.
5	Final approver	Locks and signs final MoM.	Tasks are activated and final record is archived.

Notifications

- Meeting invited/changed
- MoM ready for review
- Correction requested
- Signature required
- Action item due

Acceptance Criteria

- ☐ A contractor can comment on a MoM without accessing internal meeting notes.

CONSTRUCTION OS / MASTER PRD

- ☐ Final MoM preserves signatures and cannot be edited in place.

17. Contracts, Legal, Claims and Penalties

Centralises company, client and contractor agreements, legal obligations, variations, claims, penalties, fees and dispute workflows.

Primary actors	Legal, Finance, Project Managers, Company Owner, Contractors, Clients
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Contract profile with number, parties, value, currency, signed date, start/end, retention, warranty and status.
- Signed contract, annexes, variations, change orders, notices and correspondence.
- Milestones, payment schedule, penalties, extensions and obligations.
- Claims, disputes, court/arbitration cases, legal costs and settlements.
- Contractor right of response to penalties through restricted portal.
- Internal legal comments remain private; contractor-facing notice and evidence are explicitly shared.
- Public tender documents may be published cross-company; final signed contract remains restricted to authorised parties.

Business Rules

CON-001 Penalty response [Must]

Contractor may accept, dispute, comment, upload evidence or request a meeting.

CON-002 Legal decision [Must]

Legal/Project Management may confirm, reduce, cancel or escalate a penalty with mandatory reasoning.

CON-003 Financial linkage [Must]

Final penalty and legal fees update contractor financial overview and project cost records.

CON-004 Expiry alerts [Must]

Contracts, insurance, licenses and warranties trigger configurable reminders.

Key Workflows

Penalty Dispute

Step	Actor	Action	System result
1	Project Manager/Legal	Creates penalty with amount, reason, contract clause and evidence.	Contractor is notified.
2	Contractor	Accepts or disputes; adds comments/evidence.	Status changes to Under Review.
3	Legal/PM	Reviews and records final decision.	Penalty is confirmed, reduced, cancelled or escalated.
4	Finance	Applies final amount to balances where required.	Financial records and timelines update.

Notifications

- Contract signed/expiring
- Variation submitted/approved
- Penalty issued/responded/decided

CONSTRUCTION OS / MASTER PRD

- Legal deadline
- Warranty ending

Acceptance Criteria

- ☐ Contractor can dispute only penalties belonging to its own company.
- ☐ Legal fees and final penalties appear in financial and contractor profiles.

18. Contractor Profiles and Contractor Portal

Creates a permanent 360-degree contractor record covering contacts, qualification, contracts, payments, penalties, performance, meetings and warranty.

Primary actors	Procurement, Legal, Finance, Project Managers, QA/QC, HSE, Contractor Representatives
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Company identity, registration, VAT, licenses, trades, address, bank details and status.
- Multiple contacts with roles, phone, email, WhatsApp and language.
- Prequalification: insurance, certifications, safety, financial review, references and risk rating.
- Contracts, variations, invoices, transfers, retentions, balances, penalties, legal fees and claims.
- Project history, tasks, inspections, issues, meetings, communications and warranty obligations.
- Performance score for quality, schedule, safety, communication, documentation and payment/claim behaviour.
- Restricted portal for own records, documents, tasks, invoices, penalties and comments.

Business Rules

CON-001 Portal isolation [Must]

Contractor cannot view other contractors or unrelated company/project information.

CON-002 Qualification gate [Must]

Only approved contractors may be selected for new work unless an authorised override is recorded.

CON-003 Balanced review [Should]

Company and contractor evaluations are recorded separately; subjective ratings are secondary to objective performance.

Acceptance Criteria

- ☐ The contractor overview displays signed value, paid amount, outstanding amount, penalties and legal fees.
- ☐ A contractor portal user sees only records belonging to their contractor company.

19. Finance, Budgeting, Payments and Spending

Provides project and company financial control, including budgets, spending requests, invoices, payments, cash flow, contractor exposure and approvals.

Primary actors	Finance, Company Owner, CEO, Project Managers, All users for spending submission, Legal
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Company/project budgets, cost centres, accounts, revenue, expenses and forecasts.
- Invoices, bills, receipts, transfers, payment status and attachments.
- Anyone may submit a spending bill with project, category, amount, currency, supplier and evidence.
- Approval chain: submitter → immediate superior → department head if configured → Finance → paid/archived.
- Threshold-based approvals and Director escalation for high-value spending.
- Budget-versus-actual, cash flow, profit/loss, receivables, payables and retention.
- Contractor financial overview linked to contracts, variations, penalties and legal fees.

Business Rules

FIN-001 Two-stage spending approval [Must]

Every spending bill requires superior approval before Finance approval.

FIN-002 Evidence requirement [Must]

Receipt/invoice is required unless a policy exception is selected and approved.

FIN-003 Budget control [Must]

Over-budget or unbudgeted spending triggers extra approval.

FIN-004 Manual save [Must]

Financial forms do not autosave committed records; drafts are explicitly saved.

FIN-005 European format [Must]

Default numbers use European format 1.000,50; currency is always explicit.

Key Workflows

Spending Bill

Step	Actor	Action	System result
1	Any authorised user	Creates draft bill and uploads receipt/invoice.	Draft remains private.
2	Submitter	Submits bill.	Immediate superior is notified.
3	Superior	Approves, rejects or requests information.	Finance review starts only after approval.
4	Finance	Validates category, tax, budget and evidence; approves/rejects.	Approved bill is posted for payment.
5	Finance	Marks paid with transfer reference.	Project budget, contractor/vendor timeline and reports update.

Notifications

- Spending submitted
- Superior approval required
- Finance approval required
- Payment made
- Budget threshold exceeded
- Invoice overdue

Acceptance Criteria

- ☐ A bill cannot reach Finance before superior approval.
- ☐ A paid amount automatically updates the related project and contractor/vendor financial views.

20. Procurement, Suppliers and Purchase Orders

Controls sourcing, quotations, supplier selection, orders, deliveries and vendor performance.

Primary actors	Procurement, Finance, Project Managers, Legal, Stock Managers
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Supplier database and verified company link where available.
- RFQs, quotations, comparison tables and approval workflows.
- Purchase orders, change orders, expected deliveries and receiving status.
- Material/subcontract package linkage to project, contract and budget.
- Supplier performance for price, quality, delivery, response and claims.
- Centralised procurement option within parent groups subject to legal entity rules.

Business Rules

PRO-001 Supplier linkage [Should]

If supplier already exists on platform, company may link its verified profile instead of creating a duplicate.

PRO-002 PO approval [Must]

Purchase orders follow value and category thresholds.

PRO-003 Delivery reconciliation [Must]

Received quantity is reconciled with PO and warehouse receipt.

Acceptance Criteria

- ☐ A quotation comparison can be approved and converted to a PO.
- ☐ Warehouse receiving can identify PO quantity variances.

21. Warehouses, Stock and Material Accountability

Gives each stock/quantity manager a dedicated control panel for assigned warehouses and complete traceability of all goods movements.

Primary actors	Stock Managers, Procurement, Project Managers, Finance, Tagged recipients, Contractors
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Multiple warehouses, storage locations, shelves/bins and assigned managers.
- Material master with unit, quantity, minimum stock, project owner, supplier, batch, serial, expiry, value and photos.
- Stock In, Stock Out, transfer, return, damaged/lost, correction and reservation.
- Mandatory end-of-day prompt asking whether anything entered or left the warehouse.
- Stock manager comments where goods went and tags any OS user—architect, engineer, contractor or other authorised recipient.
- Tagged recipient confirms, partially confirms, disputes, returns or comments on receipt.
- Daily close, cycle counts, audits, variance explanations, QR/barcode scanning and reports.

Business Rules

WAR-001 Daily confirmation [Must]

Each assigned stock manager must confirm No Movement or record movements for each working day.

WAR-002 Persistent escalation [Must]

Unconfirmed daily movement remains in notifications and escalates to superior/Procurement/PM.

WAR-003 Recipient confirmation [Must]

Stock Out remains open until tagged receiver confirms or an authorised manager closes with reason.

WAR-004 Insufficient quantity [Must]

System blocks issue above available stock unless authorised override with mandatory comment.

WAR-005 Transfer dual confirmation [Must]

Sending manager confirms dispatch; receiving manager confirms receipt; differences create an issue.

Key Workflows

Tagged Stock Out

Step	Actor	Action	System result
1	Stock Manager	Selects item, quantity, project, destination, purpose and related task; adds comment.	Available stock is validated.
2	Stock Manager	Tags receiving user/contractor and submits movement.	Stock is reserved/issued; receiver is notified.
3	Receiver	Confirms quantity, disputes or reports damage with evidence.	Movement closes or issue workflow begins.
4	System	Writes event to warehouse, project, task, contractor and user timelines.	Reports and valuation update.

Notifications

- End-of-day confirmation
- Low stock
- Delivery received
- Goods issued to you
- Receipt disputed
- Inventory variance

Acceptance Criteria

- ☐ A stock manager can tag an Architect or Contractor and the recipient can confirm.
- ☐ No end-of-day response escalates automatically.

22. Architecture and Engineering Workspaces

Provides discipline-specific dashboards and workflows while maintaining shared project coordination.

Primary actors	Architects, Engineers, BIM Managers, Project Managers, Clients, Sales
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Role-specific fixed dashboards for active projects, drawings, RFIs, revisions, submittals, approvals and tasks.
- Architectural design packages, floor plans, elevations, details, material schedules and renders.
- Engineering disciplines: structural, civil, MEP, calculations, specifications and inspections.
- Coordination tasks across Architecture, Structural, MEP, QA/QC, Legal and Finance.
- Sales floor-plan requests and client customisation workflows.
- Digital signatures and approved-document status.

Business Rules

ARC-001 Role dashboard [Must]

Architects and Engineers see only relevant modules and assigned projects by default.

ARC-002 Cross-discipline approval [Must]

Hybrid sequential/parallel workflow is supported.

ARC-003 Design promise control [Must]

Sales cannot promise technical changes before required design/engineering/legal/finance approvals.

Acceptance Criteria

- ☐ An Architect receives a floor-plan request and uploads final PDF/DWG package.
- ☐ A structural reviewer can reject a facade revision without accessing unrelated Finance data.

23. Sales, CRM, Units and Client Customisation

Manages leads, clients, project units, reservations, sales packages, floor-plan requests and approved customisation from inquiry to handover.

Primary actors	Sales, Architects, Engineers, Finance, Legal, Clients, Project Managers
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Client profile, contact details, interests, budget, communication, meetings and documents.
- Unit status: available, reserved, sold, under negotiation or blocked.
- Reservations with expiry, salesperson and conditions.
- Floor-plan request tasks to Architecture/Interior/Engineering with required deliverables.
- Client customisation request: design, technical feasibility, cost, legal variation, client approval and construction.
- Sales package generator with floor plans, brochure, specifications, price, payment plan and availability.
- Sales KPIs, targets, pipeline, conversion and commissions where configured.

Business Rules

SAL-001 Reservation consistency [Must]

A unit cannot be actively reserved or sold to multiple clients.

SAL-002 Customisation approval [Must]

No client change proceeds to construction before required approvals and client acceptance.

SAL-003 Approved collateral [Must]

Sales uses only approved floor plans and materials.

Key Workflows

Client Floor Plan Request

Step	Actor	Action	System result
1	Sales	Creates task linked to client and unit.	Architecture is assigned required PDF/DWG/furniture/area package.
2	Architect	Prepares draft and submits for internal review.	Review workflow starts.
3	Approver	Approves final package.	Sales and client are notified according to permissions.

Acceptance Criteria

- ☐ Sales can see available units in real time.
- ☐ A client modification request automatically creates linked design, cost and legal actions.

24. Client Collaboration Portal

Gives developers/property owners configurable access to progress, documents, decisions and approvals without exposing internal company data.

Primary actors	Clients, Project Managers, Company Admin, Architects, Finance, Legal
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- View project progress, schedules, approved drawings, photos and reports.
- Approve drawings, variations, quotations and invoices where configured.
- Create requests, RFIs and change requests.
- Participate in meetings, MoM approval and decision voting.
- Upload documents, comment on permitted tasks and digitally sign records.
- Access Digital Twin views if subscribed and permitted.

Business Rules

CLI-001 Least privilege [Must]

Client portal permissions are project- and record-specific.

CLI-002 Internal separation [Must]

Internal comments, budgets and legal strategy remain hidden unless explicitly shared.

Acceptance Criteria

- ☐ A client can approve a drawing without seeing internal review notes.
- ☐ Client access can be revoked without deleting signed approvals.

25. Site Operations, QA/QC and HSE

Captures daily field activity, quality inspections, safety controls, evidence and corrective actions online or offline.

Primary actors	Site Managers, QA/QC, HSE, Project Managers, Contractors
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Daily site report: weather, labour, equipment, deliveries, completed work, problems, photos and next-day plan.
- QA inspections, checklists, pass/fail, non-conformities, punch lists, retests and approvals.
- HSE inspections, incidents, PPE, toolbox talks, permits, risk assessments and emergency information.
- Issue tracking with owner, priority, root cause, corrective action and closure evidence.
- Progress photos and visual timelines.
- Offline completion and later synchronisation.

Business Rules

SIT-001 Inspection evidence [Must]

Failed inspections require issue/corrective action before closure.

SIT-002 Safety escalation [Must]

Critical incidents immediately notify designated management and HSE contacts.

SIT-003 Offline traceability [Must]

Offline timestamps, device and sync time are recorded.

Acceptance Criteria

- ☐ A QA inspector can complete an inspection offline and sync later.
- ☐ A failed inspection creates a trackable corrective-action task.

26. Assets, Equipment and Fleet

Tracks company assets, machinery, vehicles, assignments, maintenance, fuel, documents and utilisation.

Primary actors	Equipment Managers, Fleet Managers, Project Managers, Finance, HSE
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Asset record, QR code, location, assigned user/project, purchase value, warranty and depreciation metadata.
- Equipment availability, operator, maintenance, fuel and reservation.
- Vehicle driver, mileage, trips, insurance, inspection and service history.
- Maintenance alerts, service tasks and downtime.
- No cross-company resource lending outside explicit JV operational rules.

Business Rules

ASS-001 Double booking [Must]

Equipment reservations cannot overlap unless override is approved.

ASS-002 Document expiry [Must]

Insurance and technical-inspection expiries generate alerts.

Acceptance Criteria

- ☐ A crane cannot be reserved for two projects at the same time.
- ☐ Maintenance history remains linked after project reassignment.

27. Approval and Workflow Engine

Standardises sequential, parallel, mixed and conditional approvals across every module.

Primary actors	Company Admin, Department Heads, Approvers, All modules
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Sequential, parallel and mixed approval stages.
- Conditional branches by amount, risk, department, contract impact, project or category.
- Delegation, temporary approver, escalation and expiry.
- Reapproval after revision and stage reset rules.
- One Approval Centre for pending items.
- Subscription-based company workflow configuration; custom modules remain platform-developed.

Business Rules

APP-001 No silent bypass [Must]

Any override requires authorised permission, reason and audit event.

APP-002 Parallel completion [Must]

Stage completion rules may require all, minimum count or selected mandatory approvers.

APP-003 Access consistency [Must]

Approver receives temporary record access only when necessary and according to policy.

Acceptance Criteria

- ☐ A facade drawing can require Architect and Structural in parallel, followed by QA/QC then Client.
- ☐ A €10,000+ spending request can add Director approval automatically.

28. Global Search, Access Requests and AI Assistant

Enables permission-aware discovery, locked-result access requests and AI-assisted company knowledge without bypassing security.

Primary actors	All users, Record Owners, Approvers, Company Admin
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Smart global search across projects, tasks, documents, contracts, companies, people, meetings, finance, stock and timelines.
- Restricted results appear locked with safe metadata only.
- User can request permanent, temporary or read-only access with mandatory reason.
- Record owner or designated approver approves/rejects with duration.
- AI distinguishes company knowledge from general knowledge and cites source records internally.
- AI can summarise progress, identify delays, search documents, draft reports and suggest assignments.
- AI can use only data the current user is authorised to access.

Business Rules

GLO-001 Locked metadata safety [Must]

Locked search results may reveal title/type but not confidential contents, values or hidden participant names if policy forbids.

GLO-002 AI permission parity [Must]

AI access must be no broader than direct user access.

GLO-003 Access request logging [Must]

Request reason, approver, outcome, scope and expiry are audited.

Key Workflows

Request Access from Search

Step	Actor	Action	System result
1	User	Clicks locked result and enters reason.	Access request is created.
2	Record owner/approver	Approves permanent, temporary, read-only or rejects.	Access policy is updated or request closes.
3	System	Notifies requester and audits the outcome.	Search result becomes accessible for approved scope/duration.

Acceptance Criteria

- ☐ AI refuses to summarise a Finance contract the Architect cannot access.
- ☐ Temporary access expires automatically.

29. Company Profiles, Business Directory, Tenders and Trust

Creates a controlled construction-business network behind the private OS, enabling verified discovery, public projects, tendering and cross-company communication.

Primary actors	Company Owners, Business Development, Procurement, Legal, Verified companies, Professionals
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Every company has a profile with logo, description, contacts, services, countries, certifications, parent/child companies and public projects.
- Company determines public, platform-only and private fields.
- Verified badges and professional Trust Score based mainly on objective platform data.
- Business Directory filters by country, trade, certification, experience, verified status and services.
- Public tender/opportunity with scope, drawings, specifications, BOQ, requirements, dates and visibility.
- Companies apply with proposal, price, programme, credentials and questions.
- Evaluation compares price, schedule, performance, safety, quality, workload and certifications.
- Award links selected company as contractor and creates collaboration records.
- Secure company messaging can become meeting, tender, task or JV invitation.

Business Rules

COM-001 Tender confidentiality [Must]

Internal estimate, evaluation notes and final contract are not public.

COM-002 Company score integrity [Must]

Trust Score uses objective data as primary input; subjective reviews are secondary and only from verified relationships.

COM-003 Parent/child visibility [Must]

Parent company profile may show child companies without exposing private child data.

Acceptance Criteria

- ☐ A company can publish a facade tender visible to verified companies and receive applications.
- ☐ The final signed contract is shared only with selected parties.

30. Digital Twin and BIM/3D Integration

Provides premium navigation and control through a live 2D/3D project model linked to Construction OS records.

Primary actors	Project Teams, Executives, Clients, Contractors, Digital Twin Admins
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Navigate project, building, block, floor, zone, unit, room and model elements.
- Select object to view progress, responsible contractor, tasks, drawings, RFIs, inspections, photos, materials, spending, contracts, risks and timeline.
- Link model GUIDs to system entities and maintain model-version mapping.
- Support IFC first; future connectors for Revit, Navisworks, Unreal Engine, Cesium and ROZARIS technology.
- Permission-aware object data and model layers.
- Premium subscription and separate performance/storage limits.

Business Rules

DIG-001 Model is navigation, not authority [Must]

The model links to authoritative OS records; it does not replace contract/document controls.

DIG-002 Version mapping [Must]

Object relationships survive model revisions or are flagged for remapping.

DIG-003 Performance [Must]

Large models require streaming, level-of-detail and background processing.

Acceptance Criteria

- ☐ Selecting a facade panel opens its assigned tasks, contractor, inspection and drawing.
- ☐ A client sees only data permitted by the client portal.

31. Dashboards, Analytics and Reports

Delivers fixed role-specific dashboards, executive oversight and exportable reports.

Primary actors	All roles, Company Admin, CEO
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Fixed dashboard layouts per user type for consistency and training.
- CEO: active projects, revenue, pending approvals, risks, project overview, finance, activity and subsidiaries.
- HR: employees, hires, leave, vacancies, attendance, birthdays and actions.
- Architect: projects, drawings, RFIs, revisions, approvals and activity.
- Finance: revenue, expenses, profit, cash, receivables, cash flow, accounts, transactions, budgets and payments.
- Company Admin: users, roles, invitations, licensed seats, security and audit actions.
- Reports: project, contract, contractor, finance, HR, stock, procurement, QA, HSE, risks, sales and daily site.
- PDF/Excel export with company branding according to subscription.

Business Rules

DAS-001 Role consistency [Must]

Users cannot rearrange dashboard widgets in V1.

DAS-002 Branding [Must]

Full branded PDF templates are subscription-dependent.

DAS-003 Status colours [Must]

Platform status colours are fixed to preserve universal meaning.

Acceptance Criteria

- ☐ Finance and Architect accounts open different dashboards from the same platform.
- ☐ PDF report uses company-approved branding without exposing platform-internal data.

32. Offline Mode and Synchronisation

Supports field work in poor-connectivity environments while maintaining security, data integrity and auditability.

Primary actors	Site staff, QA/QC, HSE, Stock Managers, All supported mobile users
Module visibility	Role- and subscription-controlled; tenant and record permissions always apply.

Functional Scope

- Offline create/update for tasks, comments, photos, inspections, stock, attendance, daily reports, MoM and signatures.
- Explicit download of drawings and project data for offline use.
- Encrypted device storage, per-user workspace isolation and remote logout policy.
- Background sync, retry queue, progress and failure visibility.
- Conflict detection and user-assisted merge for incompatible edits.
- Server is authoritative after conflict resolution; all versions remain in history.

Business Rules

OFF-001 Sensitive exclusions [Must]

Finance/legal records may be excluded from offline storage by policy.

OFF-002 Conflict safety [Must]

System must never silently overwrite a conflicting server version.

OFF-003 Sync audit [Must]

Creation time, device time, sync time and actor are recorded.

Acceptance Criteria

- ☐ A stock issue recorded offline syncs exactly once when connectivity returns.
- ☐ Two conflicting edits produce a merge/review state rather than data loss.

33. UX/UI and Responsive Design Specification

- Light mode only.
- Flat, premium, minimalist visual language with dark-grey typography and restrained gold/brand accents.
- Subtle 150–250 ms transitions and high-quality toggle/on-off animation.
- High information density managed through clear hierarchy, cards, tabs, tables, drawers and progressive disclosure.
- Desktop: collapsible sidebar, top search/notifications, main content and optional right activity/approval region.
- Tablet: two-column layouts, collapsible panels and simplified tables.
- Mobile: single-column summaries, card-based table rows, bottom/sheet navigation, full-width forms and tap-based help.
- Sidebar auto-collapses on smaller screens.
- Skeleton loading with subtle branded animation; interactive empty states.
- No keyboard shortcuts in V1.
- Favorites and Recently Opened are supported.
- Calendar default is user-selected; time format is user choice with 12-hour default; number format is European.
- Session restoration: ask user whether to restore previous workspace.

Contextual “?” Help System

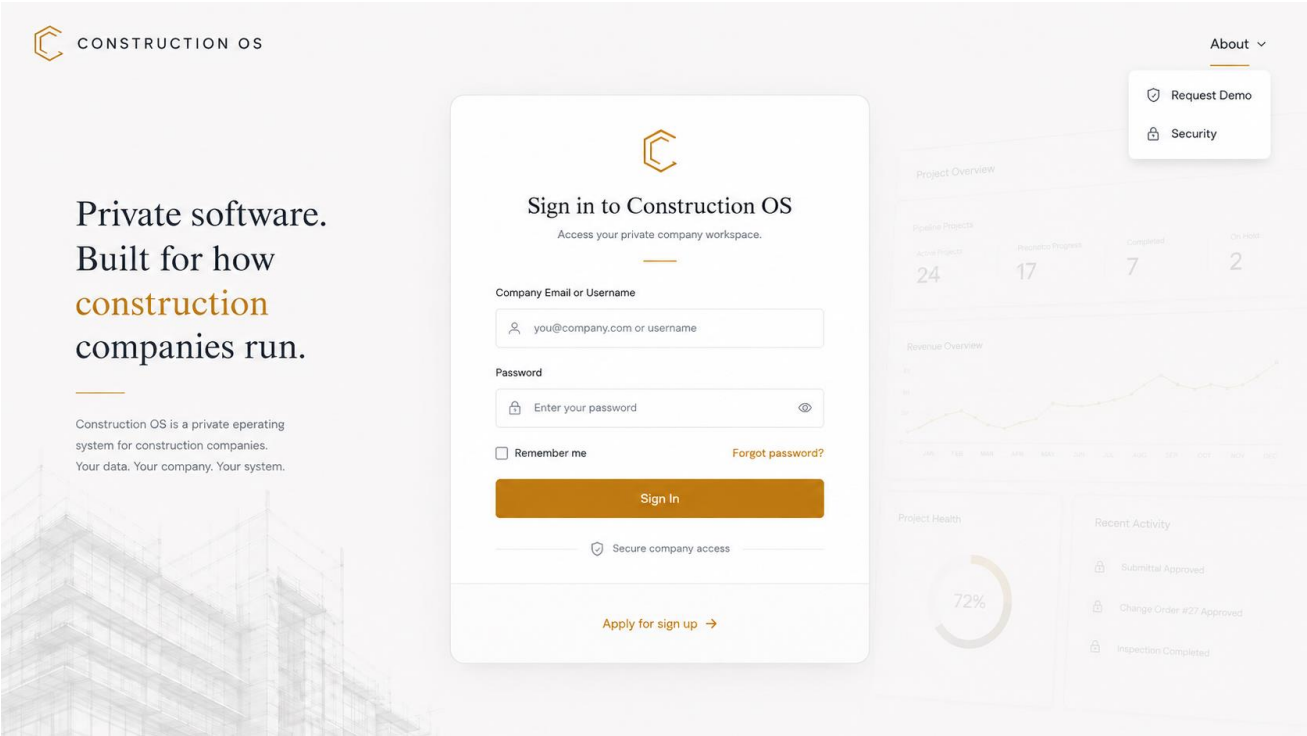
- A minimal question-mark icon appears after important labels, fields, statuses and features.
- Desktop: hover/focus tooltip. Mobile/tablet: tap opens a popover or bottom sheet.
- Help explains meaning, why it matters, expected data, format, example, responsibility and validation.
- Complex help may link to internal documentation or “Learn more”.
- Tooltips must support keyboard focus, screen readers and sufficient dismissal time.

Landing Page

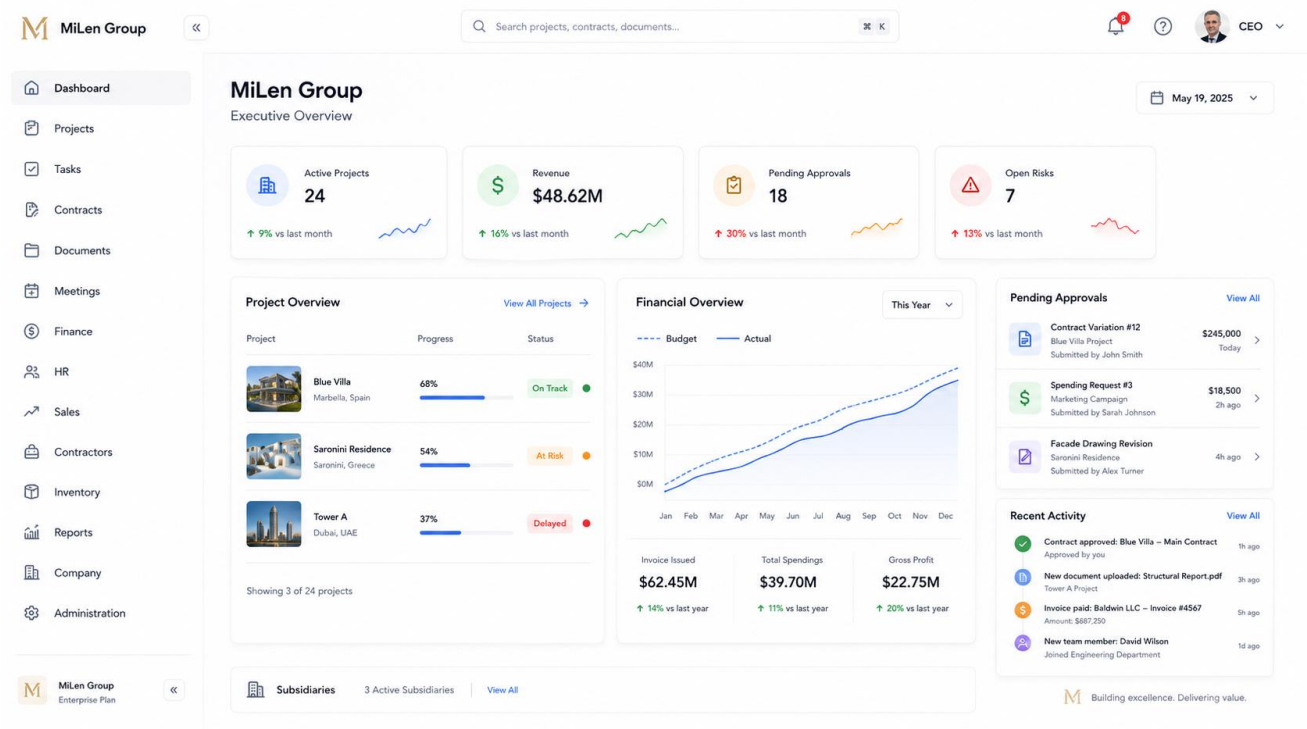
- Login-first page with central sign-in card for company username/password.
- Single understated “Apply for sign up” link for professionals or contractors who need a platform profile.
- About menu contains Request Demo and Security.
- No visible three-platform narrative; the message is private software for construction companies.
- Fully responsive with login card prioritised on mobile.

34. Approved Visual References

These renders define direction, not pixel-perfect production layouts. The development design system must normalise branding, typography, spacing, data density and accessibility across all screens.



Landing Page — final reference: central login and Apply for sign up.



CEO Login / Executive Dashboard — simplified role-specific reference.

Construction OS

Company Admin

Dashboard

USER MANAGEMENT

Users

Roles & Permissions

Architects

Teams

Invitations

COMPANY SETTINGS

Company Profile

Subscription

Integrations

Security

Audit Logs

Company Admin Dashboard

Manage your team, roles and access to Construction OS.

Total Users

24

↑ 5 this month

Architects

8

↑ 2 this month

Active Projects

12

No change

Pending Invitations

3

[View invites](#)

Users

Manage and assign access to your team members.

Filter

Create User

User	Role	Team	Status	Last Active	Actions
 Arben Kola arben.kola@buildcore.com	Company Admin	Management	Active	May 20, 2025	
 Elira Doda elira.doda@buildcore.com	Architect	Design Team	Active	May 19, 2025	
 Gentian Hoxha gentian.hoxha@buildcore.com	Project Manager	Projects	Active	May 18, 2025	
 Sara Mema sara.mema@buildcore.com	Architect	Design Team	Active	May 17, 2025	
 Besnik Lala besnik.lala@buildcore.com	Site Manager	Construction	Active	May 16, 2025	

[View all users](#)

Pending Invitations

Invitations waiting for acceptance.

[View all](#)

NM	new.architect@buildcore.com Architect	Sent May 20, 2025	Pending
PM	pm@buildcore.com Project Manager	Sent May 19, 2025	Pending
SM	site.manager@buildcore.com Site Manager	Sent May 18, 2025	Pending

[Resend Reminder](#)

Quick Actions

Create New User
Add a new team member

Assign Architect
Grant architect access to Construction OS

Manage Roles
Configure roles and permissions

View All Teams
Manage teams and their members

Roles Overview



[Manage roles & permissions](#)

Recent Activity

[View all](#)

New user created

 Elira Doda was added as Architect
May 20, 2025, 10:30 AM

Role updated

 Besnik Lala role changed to Site Manager
May 19, 2025, 04:45 PM

Access revoked

 User Iir Meta access was revoked
May 18, 2025, 11:20 AM

Invitation sent

 Invitation sent to new.arch@buildcore.com
May 17, 2025, 09:15 AM

[View all activity](#)

Current Plan

Enterprise

Next billing: Jun 1, 2025

23 / 50 Users

[Manage Subscription](#)

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Company Admin Dashboard — user creation, roles, invitations and access management.

Construction OS

HR Manager

Human Resources

Overview

Employees

Recruitment

Onboarding

Attendance

Leave Requests

Documents

Training

Performance

Reports

Settings

HR Dashboard

Welcome back! Here's what's happening with your team.

Total Employees

128

↑ 8 this month

New Hires

7

↑ 2 this month

On Leave

12

— No change

Open Positions

5

↑ 1 this week

Recent Hires

[View all](#)

 Arben Kola Site Engineer	May 20, 2025
 Elira Doda Project Coordinator	May 18, 2025
 Gentian Hoxha Quantity Surveyor	May 15, 2025
 Besnik Lala Safety Officer	May 12, 2025
 Sara Mema Architect	May 10, 2025

[View all new hires](#)

Leave Requests

[View all](#)

 Ardit Nuredini Jun 2 – Jun 6, 2025	Pending
 Leonard Tafa Jun 5 – Jun 7, 2025	Pending
 Drenisa Marku Jun 9 – Jun 13, 2025	Approved
 Jetmir Pllana May 28 – May 30, 2025	Approved
 Klodian Shala May 27 – May 29, 2025	Rejected

Employee Distribution



Upcoming Birthdays

[View all](#)

 Elira Doda May 28	Tomorrow	
 Gentian Hoxha Jun 2	In 6 days	
 Sara Mema Jun 5	In 9 days	

Quick Actions

[Add Employee](#)
Create a new employee profile

[Request Report](#)
Generate HR reports

[Add Document](#)
Upload policies or documents

[Create Announcement](#)
Send news to the team

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 **Construction OS**
BuildCore Group

Overview

DESIGN MANAGEMENT

Projects

Drawings

Revisions

RFIs

Submittals

Approvals

Tasks

Meetings

Documents

PROJECT SETTINGS

Settings

Current Plan

Enterprise

Next Billing: Jun 1, 2025

23 / 50 Users

Manage Subscription

Architect Dashboard

Welcome back, Elira. Here's what's happening with your design projects.

Active Projects

12

↑ 2 this month

Pending Drawings

18

↑ 4 this week

Open RFIs

9

↓ 1 this week

Revisions Awaiting Approval

6

No change

Drawing Packages

Track and manage drawing sets and revisions.

Search drawings...

Filter

Upload

Project	Package / Set	Revision	Status	Updated	Actions
 Riverside Cultural Center RCC-2104	A1.0 — Overall Plans	Rev. 3	In Review	May 20, 2025	
 Harborview Office Tower HOT-2402	A2.3 — Floor Plans	Rev. 2	Approved	May 19, 2025	
 Maplewood Mixed-Use MMU-2207	A5.1 — Elevations	Rev. 4	Needs Revision	May 18, 2025	
 Lakeview Community Center LCC-2301	A0.1 — Cover Sheet	Rev. 1	Draft	May 17, 2025	

View all drawing packages →

Design Package Overview

Approved

38%

In Review

32%

Needs Revision

15%

Draft

15%

Total Packages

53

View all packages →

Recent RFIs

View all

RFI

RFI-023 | Stair Guard Height

Riverside Cultural Center

May 20, 2025

Open

RFI

RFI-022 | Window Type Clarification

Harborview Office Tower

May 19, 2025

Answered

RFI

RFI-021 | Roof Drain Locations

Maplewood Mixed-Use

May 18, 2025

Open

RFI

RFI-020 | Door Hardware Specs

Lakeview Community Center

May 17, 2025

Answered

View all RFIs →

Upcoming Approvals & Deadlines

View all

Building Elevations — Rev. 3

Riverside Cultural Center

May 22, 2025

Approval

Interior Finish Package

Harborview Office Tower

May 23, 2025

Approval

Design Development Set

Maplewood Mixed-Use

May 26, 2025

Deadline

MEP Coordination Set

Riverside Cultural Center

May 27, 2025

Approval

View all approvals →

Recent Activity

View all

Drawing uploaded

A2.3 — Floor Plans — Rev. 2

Harborview Office Tower

10:12 AM

Revision approved

A1.0 — Overall Plans — Rev. 2

Riverside Cultural Center

Yesterday

RFI answered

RFI-019 — Ceiling Height Clarification

Maplewood Mixed-Use

May 19, 2025

Submittal approved

Glass & Glazing — Submittal 12

Harborview Office Tower

May 18, 2025

View all activity →

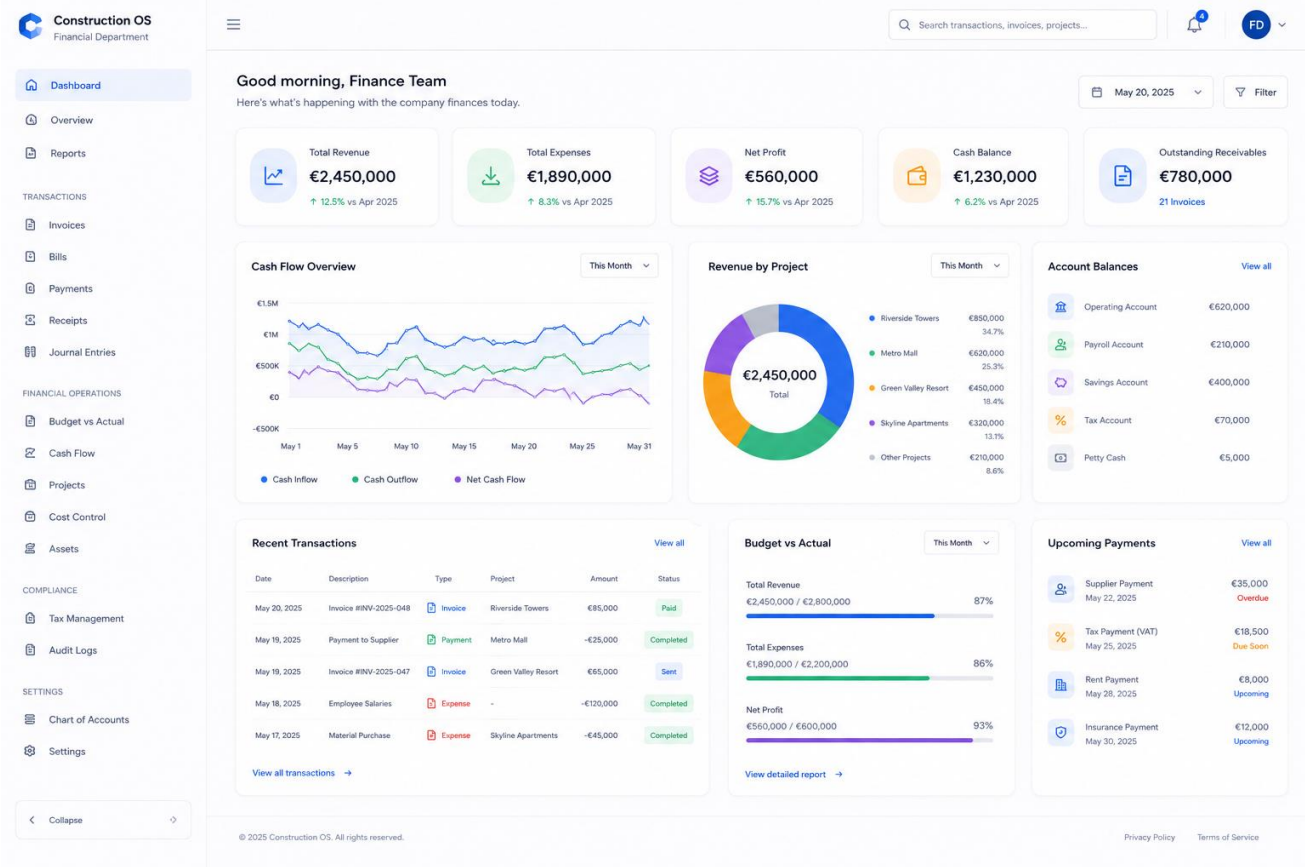
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Architect Dashboard — drawings, RFIs, revisions and approvals.

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Finance Dashboard — finance department overview.

35. Recommended Technical Architecture

The final stack may be changed by the senior engineering team after an architecture review, but the following decomposition is the required baseline for estimation.

Frontend	TypeScript, React/Next.js or equivalent enterprise web framework; responsive PWA and offline-capable service worker.
Backend	Modular TypeScript/.NET/Java service architecture with clear domain boundaries; start as modular monolith unless scale justifies services.
API	Versioned REST/JSON APIs documented with OpenAPI; real-time events via WebSocket/SSE where appropriate.
Database	PostgreSQL recommended; tenant-scoped schema/data model with dedicated-database deployment option.
Files	Object storage with tenant-prefixed keys, signed URLs, malware scanning, metadata database and retention policies.
Search	Permission-aware search index (OpenSearch/Elasticsearch or equivalent) with tenant and ACL filters.
Background jobs	Queue-based processing for notifications, reports, file conversion, sync, AI indexing and model processing.
Cache	Tenant-aware Redis or equivalent; never cache cross-tenant data without scoped keys.
Identity	Internal username/password authentication with secure session/token management and future-ready identity abstraction.
Observability	Central logging, metrics, traces, audit-event monitoring and security alerts.
Cloud	Containerised deployment with separate development, staging and production environments; infrastructure as code and automated CI/CD.

Domain Boundaries

- Identity and Access
- Tenant/Company and Subscription
- Projects and Work Management
- Documents and Approvals
- People/HR/Recruitment
- Contracts/Legal/Contractors
- Finance/Procurement
- Inventory/Assets
- Sales/Clients
- Network/Tenders
- Notifications/Timeline/Search
- Digital Twin

- Reporting/AI
- Audit/Security/Support

Architecture Rule

Shared cross-cutting engines—permissions, approvals, timelines, comments, notifications, files and audit—must be reusable platform services. Individual modules must not implement incompatible versions of these behaviours.

36. Core Data Model and Entity Relationships

Entity Catalogue

Entity	Purpose
Tenant	Top-level data-isolation boundary for a company workspace.
Company	Business identity; can be parent, child or independent.
Branch	Office/operating location within a company.
UserIdentity	Platform login identity.
ProfessionalProfile	Portable Construction Passport owned by individual.
CompanyMembership	Links identity to company role, position, department and status.
Role / Permission / Policy	Authorisation model and overrides.
Subscription / Entitlement	Licensed seats, modules, storage and options.
JVWorkspace / JVParticipant	Controlled collaboration between companies.
Project / ProjectMember	Project scope and memberships.
Task / TaskContribution / TaskDependency	Collaborative work and deliverables.
Decision / Vote	Advisory participation and final authority.
Document / Version / Revision / Signature	Controlled files and approvals.
RFI / Submittal	Technical questions and material submissions.
Meeting / MoM / ActionItem	Meeting record and follow-up work.
Contract / Variation / Claim / Penalty	Commercial/legal record.
Contractor / ContractorContact / Qualification	External company relationship.
Invoice / Payment / Expense / Budget	Financial records.
Supplier / RFQ / Quote / PurchaseOrder	Procurement.
Warehouse / StockItem / StockMovement	Inventory.
Employee / Attendance / Leave / Certification	HR.
Client / Unit / Reservation / SalesOpportunity	Sales.
Inspection / Issue / Risk / Incident	QA/QC/HSE.
Asset / Equipment / Vehicle / Maintenance	Physical resources.
Tender / TenderApplication	Cross-company opportunity.
CompanyProfile / Verification / TrustMetric	Business network identity.
Notification / DeliveryPreference	Communication.
TimelineEvent / Comment / Relationship	Universal history and collaboration.
ApprovalWorkflow / ApprovalInstance / ApprovalDecision	Reusable approvals.
AccessRequest / AccessGrant	Locked record request workflow.
AuditEvent / SupportSession	Security and compliance evidence.
Model / ModelObject / ObjectLink	Digital Twin.

37. Identifier, Status and Draft Standards

Permanent internal ID	UUID/ULID; never exposed as sole business reference.
Human-readable code	Tenant-configurable prefix + year + sequence, e.g., TSK-2026-000582.
Draft lifecycle	Draft → Submitted → Under Review → Approved / Rejected → Published or Completed.
Task lifecycle	To Do → In Progress → Review → Approved → Completed; alternatives include Blocked, On Hold, Needs Revision, Rejected, Overdue.
Receipt lifecycle	Awaiting confirmation → Confirmed / Partially received / Disputed / Returned → Closed.
Approval lifecycle	Pending → Approved / Rejected / Changes requested / Withdrawn / Expired.

Status	Meaning	Transition rule
Draft	Visible only to author and selected draft collaborators.	Submit to enter formal workflow.
Submitted	Formally delivered for review.	Moves to review/approval according to workflow.
Under Review	Reviewers are evaluating.	Approve, reject or request revision.
Approved	Accepted by required authority.	May publish, execute or complete.
Rejected	Not accepted.	Return to draft/revision with reason.
Published / Completed	Official or closed state.	Changes require new revision or reopen authority.

38. API and Event Requirements

The development team must deliver an OpenAPI specification covering all endpoints. The following resource groups are mandatory.

Resource group	Required coverage
/auth	login, logout, refresh/session, password change/reset
/companies	profiles, hierarchy, branches, verification, settings
/memberships	users, roles, positions, departments, invitations, access mode
/projects	projects, members, packages, progress, archive, templates
/tasks	tasks, contributions, decisions, votes, join/follow, delay records
/documents	files, versions, revisions, signatures, annotations
/rfis and /submittals	technical workflows
/meetings	meetings, MoM, action items, signatures
/contracts	contracts, variations, claims, penalties, warranties
/contractors	profiles, contacts, qualifications, performance
/finance	budgets, expenses, invoices, payments, reports
/procurement	suppliers, RFQs, quotes, POs, deliveries
/inventory	warehouses, items, movements, counts, confirmations
/hr	employees, attendance, leave, recruitment, certifications
/sales	clients, units, reservations, floor-plan/customisation requests
/quality and /hse	inspections, issues, incidents, checklists
/assets	equipment, fleet, reservations, maintenance
/network	company directory, tenders, applications, messaging
/approvals	workflow definitions, instances, decisions, delegation
/search	global search, locked metadata and access requests
/notifications	feed, preferences, delivery and acknowledgement
/timeline and /comments	events, threads, relationships
/offline-sync	change sets, conflicts and retries
/digital-twin	models, objects, links and viewer tokens
/reports	generation, status, download and schedules
/admin	subscriptions, support access, deletion and audit

API Rules

- Every request is authenticated, tenant-scoped and permission-checked.
- List endpoints support pagination, filtering, sorting and safe search.
- Write operations support idempotency where duplicate submission is dangerous.
- All errors return stable codes, human-readable messages and correlation IDs.
- Long jobs use asynchronous status resources and notifications.
- Webhooks/events must be signed, replay-safe and tenant-scoped.

39. Notification Event Matrix

Event	Recipients	Channels / policy
Task assigned	Assignee, creator, watchers	In-app; user-selected email/WhatsApp
Task completed	All included task members	In-app + selected channels
Task overdue	Assignee, task manager, escalation chain	Critical per policy
Delay explanation missing	Assignee and superior	Escalating
Drawing submitted/approved/rejected	Author, reviewers, linked task members	Selected channels
Spending submitted	Immediate superior	In-app + selected channels
Spending approved by superior	Finance	In-app + selected channels
Payment made	Submitter, finance stakeholders, linked contractor/vendor	Selected channels
Penalty issued	Contractor authorised contacts, Legal, PM	In-app/email/WhatsApp according to legal policy
Contractor response	Legal, PM, Finance if financial	In-app + selected channels
MoM ready	Attendees/reviewers	In-app + selected channels
Stock out tagged	Recipient, stock manager	In-app + selected channels
Daily stock confirmation missing	Stock manager → superior → Procurement/PM	Persistent escalation
Access requested	Record owner/designated approver	In-app + selected channels
Access granted/denied	Requester	In-app + selected channels
Certification/contract expiry	Owner, responsible department	Configurable lead times
Critical HSE incident	HSE, PM, Director, emergency contacts	Mandatory critical channels
Company deletion requested	Owner, Admins, Platform Main Admin	Email + in-app; WhatsApp optional

40. File, Storage and Document Security Requirements

- Object storage must be private by default; downloads use short-lived signed URLs and permission checks.
- Virus/malware scan every upload before availability.
- Store checksum, MIME type, extension, size, author, tenant, project, object relation, version and retention class.
- Configurable file-size limits by plan and file type.
- Support common construction formats: PDF, DWG, DXF, RVT, IFC, XLSX, DOCX, images, videos and approved archives.
- Preview generation for safe formats; source files remain unmodified.
- Digital signatures and approval stamps must be verifiable and tied to version hash.
- Watermarks and branded exports follow company subscription and permissions.
- Archive export includes manifest, metadata, timelines, approved files and audit summaries, filtered by requester permissions.
- External cloud connection imports/links content; Construction OS retains authority over official final versions.

41. Security, Privacy and Compliance

ID	Requirement	Specification
SEC-001	Password security	Use modern adaptive hashing, password-strength policy, secure reset tokens and no plaintext credential exposure.
SEC-002	Session security	Secure cookies/tokens, inactivity timeout, device/session list, revocation and CSRF/XSS protections.
SEC-003	Authorisation	Deny by default; evaluate tenant, workspace, module, project, record, role and temporary grants.
SEC-004	Encryption	TLS in transit; encryption at rest for databases, files, backups and offline device data.
SEC-005	Audit	Immutable audit events for authentication, permissions, approvals, finance, legal, support, exports and deletion.
SEC-006	Support sessions	Company-approved, scoped, time-limited, prominently visible and fully logged.
SEC-007	Data privacy	GDPR-ready rights, lawful retention, export, deletion and processor/subprocessor documentation.
SEC-008	Backups	Encrypted backups, tested restoration, documented RPO/RTO and dedicated-database procedures.
SEC-009	Search and AI	Indexes and AI retrieval must enforce the same permissions as source records.
SEC-010	Offline	Encrypted cache, device binding, remote revoke and policy-based data expiry.
SEC-011	Company deletion	Seven-day standby, owner cancellation and Platform Main Admin approval.
SEC-012	Security testing	SAST, dependency scanning, DAST, penetration testing and tenant-isolation test suite before launch.

42. Non-Functional Requirements

Non-Functional Targets

Area	Requirement
Availability	Target 99.9% monthly for standard plans; enterprise SLA separately contracted.
Performance	Typical authenticated page initial load ≤ 2.5 s on supported broadband; key API reads p95 ≤ 500 ms excluding heavy reports/files.
Search	Permission-filtered common search response p95 ≤ 2 s.
Notifications	In-app event creation ≤ 10 s; external channels subject to provider SLAs.
Scalability	Horizontal scaling for web/API/jobs; no tenant-specific code forks.
Concurrency	Architecture must support at least thousands of concurrent users and future regional expansion; final load targets established in inception.
Documents	Resumable large uploads and background preview processing.
Offline	Queue survives app restart; sync is idempotent and observable.
Browsers	Current and previous major versions of Chrome, Edge, Safari and Firefox; responsive mobile browsers.
Accessibility	WCAG 2.1 AA target for core workflows.
Localisation	English and Albanian resource files; date/time/currency formatting separated from content.
Audit retention	Configurable by legal plan and jurisdiction; critical records may be immutable for contracted period.
Observability	Structured logs, metrics, traces, tenant-safe diagnostics and alerting.
Disaster recovery	Documented RPO/RTO, tested restoration and incident runbooks.

43. QA, Testing and Acceptance Strategy

- Unit tests for domain rules, calculations, status transitions and permissions.
- Integration tests for database, files, queues, search, WhatsApp/email and background jobs.
- Contract/API tests generated from OpenAPI.
- Automated tenant-isolation tests across every resource and search path.
- Role/permission matrix tests for all roles and overrides.
- Workflow tests for tasks, delay explanations, spending, stock, penalties, MoM, approvals and JV sharing.
- Responsive UI testing on desktop, tablet and mobile breakpoints.
- Offline/sync conflict and retry tests.
- File-format, upload, virus scan, version and archive tests.
- Security testing, penetration testing and dependency scanning.
- Performance/load tests for dashboards, search, uploads, notifications and large project lists.
- User acceptance testing with CEO, Admin, Architect, HR, Finance, Project Manager, Stock Manager, Contractor and Client personas.

Definition of Done

- Acceptance criteria pass in staging.
- Permission and tenant-isolation tests pass.
- Responsive and accessibility checks pass.
- Audit events and notifications verified.
- Documentation and admin configuration supplied.
- Monitoring and rollback plan active.
- Product Owner approves workflow and visual behaviour.

44. Recommended Delivery Roadmap

Phased Roadmap

Phase	Scope
Phase 0 — Inception	Architecture validation, domain model, design system, security model, prototype, backlog, estimates and deployment foundation.
Phase 1 — Core Private OS	Authentication, tenant/company, users/roles, projects, tasks, comments, timelines, notifications, documents, approvals, search, dashboards and audit.
Phase 2 — Operations	HR, attendance, meetings/MoM, contracts, contractor profiles/portal, finance/spending, procurement, warehouses/stock.
Phase 3 — Construction Delivery	Drawings/RFIs/submittals, architecture/engineering workflows, daily site, QA/QC, HSE, assets/fleet and offline mode.
Phase 4 — Commercial and External Portals	Sales/CRM/units, floor plans, customisation, client portal, tenders, company profiles and professional recruitment.
Phase 5 — Intelligence and Premium	Advanced analytics, AI company knowledge, dedicated databases, Digital Twin, model integrations and enterprise automation.

45. Required Response from Senior Full-Stack Developer Team

- Recommended architecture and justified changes to this baseline.
- Team composition by role and phase.
- Work breakdown and estimated person-days/hours per module.
- Calendar timeline with dependencies and parallel workstreams.
- Cost per phase, ongoing infrastructure and third-party costs.
- MVP scope recommendation and explicit deferred features.
- Security, privacy and compliance approach.
- Multi-tenancy and dedicated-database migration design.
- Offline architecture and limitations.
- Search/AI permission enforcement design.
- File storage and large BIM/model handling plan.
- Testing strategy, release process, SLA and maintenance model.
- Assumptions, exclusions, risks and open questions.

Mandatory Estimation Format

Workstream	Scope	Team	Effort	Dependencies	Risk / assumption

46. Open Technical and Commercial Decisions

Open Decisions

Decision	Required outcome
Hosting region(s)	Select launch cloud/provider and primary data region.
SLA/RPO/RTO	Confirm plan-specific targets and enterprise contract options.
Payment provider	Select subscription billing/payment processor.
Email/WhatsApp providers	Select providers and country availability.
Digital signature legality	Select provider and supported jurisdictions.
Accounting integrations	Define V1 export or direct integrations by target market.
File limits	Confirm plan-specific storage and upload limits.
AI provider and quotas	Define model provider, isolation, retention, cost and plan limits.
Public profile moderation	Define publication review and prohibited content policy.
Trust Score formula	Create transparent metric model and dispute/appeal process.
Data retention	Define legal retention by record type and plan.
Professional sign-up verification	Define identity, credential and portfolio verification process.
Mobile packaging	Confirm whether PWA only or native wrappers are required after V1.
MVP countries	English/Albanian interface confirmed; confirm legal/tax localisation beyond Albania.

47. Glossary

Term	Definition
Construction OS	The complete private SaaS operating platform.
Company Owner	Highest company role with ownership, billing and deletion authority.
Company Admin	Operational administrator who creates users and configures access.
Platform Main Admin	Construction OS provider administrator.
Construction Passport	Portable professional profile and verified career identity.
JV Workspace	Cross-company collaborative project boundary.
Record owner	Person or role responsible for access and record lifecycle.
Contribution	A required deliverable assigned to a person/department inside a task.
MoM	Minutes of Meeting.
RFI	Request for Information.
Submittal	Technical/material document package submitted for approval.
Digital Twin	3D/2D model-linked operational view.
Tenant	Company data-isolation boundary.
Locked result	Search result visible by title/type but unavailable until access is granted.
Standby deletion	Seven-day read-only/cooling-off state before final deletion approval.

48. Product Baseline Sign-Off

This PRD represents the consolidated baseline for technical estimation and detailed solution design. The engineering team may recommend implementation changes, but any change to confirmed product behaviour, security, tenant isolation, permissions or legal accountability requires documented Product Owner approval.

Role	Name	Approval / Date
Product Owner		
Technical Lead		
UX/UI Lead		